

1. Overview: How Wireless MQTT Telemetry Works

SignSpeaker provides an untethered **Wireless MQTT (Message Queuing Telemetry Transport)** mode that replaces the USB cable connection. The smart glove operates as an independent IoT transmitter, streaming real-time gyroscope data over 2.4 GHz Wi-Fi to an MQTT broker, which routes packets directly to Node-RED and the Web Dashboard.



2. Hardware Wiring & Pinouts

Device	Component Pin	ESP32 GPIO Pin	Electrical Function & Notes
Glove Unit (Transmitter) ESP32 + MPU6050	VCC	3.3V	Regulated 3.3V power supply
	GND	GND	Common Ground reference
	SDA	GPIO 23	I2C Serial Data line
	SCL	GPIO 22	I2C Serial Clock line (MPU6050 address: 0x68)
Receiver Station (Optional) ESP32 + 16x2 LCD	VCC	5V (VIN)	5V supply for LCD backlight & logic
	GND	GND	Common Ground reference
	SDA	GPIO 21	I2C Serial Data line (PCF8574 address: 0x27)
	SCL	GPIO 22	I2C Serial Clock line

💡 Powering Untethered

Power the Glove ESP32 using a small portable USB power bank (5V into the micro-USB/Type-C port) or a 3.7V rechargeable LiPo battery connected to 3.3V/GND via a small regulator.

3. Arduino IDE Preparation & Required Libraries

To compile the wireless firmware sketches, install the following standard libraries in Arduino IDE:

- **PubSubClient** by Nick O'Leary (Standard MQTT client for Arduino & ESP32)
- **Adafruit MPU6050** & **Adafruit Unified Sensor** by Adafruit
- **LiquidCrystal_I2C** by Frank de Brabander (Required only for the hardware LCD receiver)

4. Flashing the Glove ESP32 (Wireless Sender)

The glove firmware is located at: `firmware/mqtt_sender/mqtt_sender.ino`

1 Open the Sketch: In Arduino IDE, open:

`d:\DATA-1\Projects\SignSpeaker Project\firmware\mqtt_sender\mqtt_sender.ino`

2 Configure Wi-Fi & Broker: Update lines 21–25 with your Wi-Fi credentials:

```
// ----- Wi-Fi & MQTT Configuration -----
const char* WIFI_SSID    = "YOUR_WIFI_NAME";      // Your 2.4 GHz Wi-Fi or Phone Hotspot name
const char* WIFI_PASS    = "YOUR_WIFI_PASSWORD";  // Your Wi-Fi password

const char* MQTT_BROKER  = "test.mosquitto.org";   // Public cloud broker (or broker.hivemq.com)
const int   MQTT_PORT    = 1883;                  // Standard MQTT unencrypted port
const char* MQTT_TOPIC   = "signspeaker/glove/data";
```

Wi-Fi Band Requirement (2.4 GHz)

ESP32 microcontrollers only connect to **2.4 GHz** Wi-Fi networks. If you are using a mobile phone hotspot, make sure to enable **"Maximize Compatibility"** (iOS) or select **"2.4 GHz AP Band"** (Android).

3 Upload to Board:

- Connect the Glove ESP32 to your PC via USB cable.
- Select **Tools → Board → ESP32 Arduino → ESP32 Dev Module**.
- Select the active COM port under **Tools → Port**.
- Click **Upload** (`Ctrl + U`).

4 Verify in Serial Monitor: Open Serial Monitor at **115200 baud**. You will see:

```
MPU6050 ready
Connecting to Wi-Fi: MyHotspot
Wi-Fi connected! IP: 192.168.43.112
Connecting to Mosquitto MQTT: test.mosquitto.org
Connected to Mosquitto broker!
D,1040,0.012,-0.005,0.021
D,1080,0.015,-0.004,0.022
```

5 Go Wireless: Unplug the USB cable from the PC! Connect the ESP32 to your battery or portable power bank. The glove is now completely wireless.

5. Flashing the Wireless 16×2 LCD Receiver (Optional)

The wireless receiver sketch is located at: `firmware/mqtt_receiver/mqtt_receiver.ino`

- Open `mqtt_receiver.ino` in Arduino IDE.
- Set the exact same `WIFI_SSID`, `WIFI_PASS`, and `MQTT_BROKER` as the sender.
- Upload to the second ESP32.
- Plug into any standard 5V USB phone charger across the room. The LCD will boot and display:

```
SignSpeaker
Wireless MQTT
```

6. Connecting the Web Dashboard

1 **Open the Web Dashboard:** Ensure Node-RED is running, then open Chrome or Edge to:
`http://127.0.0.1:1880/glove`

2 **Select Wireless MQTT Mode:** In the top header bar, click the  **MQTT** toggle button.

3 **Inspect / Change Broker:** Click  **Config** to inspect or adjust settings:

Field	Recommended Value	Description
Broker Host	<code>test.mosquitto.org</code>	Public Eclipse Mosquitto broker (or <code>broker.hivemq.com</code>)
Broker Port	<code>1883</code>	Standard unencrypted MQTT port
Data Topic	<code>signspeaker/glove/data</code>	Inbound telemetry stream from glove
LCD Topic	<code>signspeaker/glove/lcd</code>	Outbound commands to hardware LCD

4 **Click Connect:** Click the prominent blue  **Connect** button in the top right header.

5 **Confirmation Checklist:**

- Status pill turns **Green**:  Mosquitto • ~25 Hz
- The **Live Sensor** canvas chart immediately starts plotting real-time gyro traces.
- In **Train** mode: record gesture samples wirelessly.
- In **Recognize** mode: gestures trigger instant speech audio and LCD display updates!

7. MQTT Telemetry Protocol Reference

Topic	Direction	Format	Example Payload & Description
<code>signspeaker/glove/data</code>	Glove → Broker	<code>D,<ms>,<gx>,<gy>,<gz></code>	<code>D,14250,0.125,-0.045,0.892</code> Timestamp in ms, angular velocity in rad/s
<code>signspeaker/glove/lcd</code>	Node-RED → Receiver	<code>W,<word></code>	<code>W>Hello</code> Displays detected word for 2.5 seconds
<code>signspeaker/glove/lcd</code>	Node-RED → Receiver	<code>L,<line1> <line2></code>	<code>L,SignSpeaker Ready Listening...</code> Updates idle status screen

8. Troubleshooting Guide

Symptom	Probable Cause	Solution
Serial monitor prints dots <code>....</code> and never connects to Wi-Fi	SSID/Password mismatch or 5 GHz Wi-Fi	Verify Wi-Fi name & password. Ensure router/hotspot operates on 2.4 GHz band.
Glove connects to Wi-Fi, but prints <code>Failed (state -2)</code>	Broker hostname or port blocked	Ensure PC has internet access. Alternatively, switch broker to <code>broker.hivemq.com</code> in sketch and dashboard.
Dashboard status says MQTT offline	Broker configuration or network issue	Click  Config , ensure broker is <code>test.mosquitto.org</code> , and click Save & Connect .
Serial monitor says <code>MPU6050 not found</code>	Loose jumper wire or wrong pins	Check I2C wiring: SDA must be on GPIO 23 , SCL on GPIO 22 , VCC on 3.3V .